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Heterogeneous silicon-on-lithium niobate electro-optic modulator for 100-Gbaud modulation

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ABSTRACT

Integrated lithium niobate (LN) electro-optic (EO) modulators are emerging for applications in next-generation optical fiber communication networks. To date, LN crystal waveguides have led the technology for high-speed modulators. On the other hand, on-chip LN modulators are expected to realize scalable signaling devices with mature complementary metal–oxide–semiconductor technology. In this study, a silicon-loaded LN modulator on the insulator substrate featuring a small footprint, a low driving voltage, and high-speed EO modulation is designed and fabricated. No etching or patterning of the LN is required. The measured halfwave-voltage length product is 1.9 V cm with a static modulation extinction ratio of 17.9 dB. The fabricated LN modulator has a modulation bandwidth of 60 GHz and supports high-speed signaling at a data rate up to 200 Gbit/s.

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I. INTRODUCTION

As critical components of optoelectronic interconnections, electro-optic (EO) modulators have attracted much attention in applications such as short-reach optical interconnects, long-haul fiber communications, and on-chip optical links.^{1–3} Over the past several decades, EO modulators based on lithium niobate (LN) crystals have been explored in industrial applications because of their relatively large Pockels coefficient (r_{33}) of 31 pm/V, wide optically transparent window of 0.4 to 5.0 μm ,⁴ and high Curie temperature.⁵ Thus, LN modulators, which can be prepared using a titanium diffusion technique (Ti: LiNbO₃), have been realized with various waveguide structures. Previously developed LN modulators have consisted of relatively long phase shifters (longer than several tens of millimeters) and additional sections of gently curved waveguide structures.⁶ The modal confinement of the LN waveguide is attributed to the refractive indices of the Ti-diffused and non-diffused waveguide regions. However, because of the small refractive index difference of around 10^{-3} ,⁷ the relatively weak

modal confinement around the core makes it difficult to reduce the modulator drive voltage and for the EO performance of the intrinsic material to be evaluated. The challenge in developing a small-footprint LN-based device is to achieve a highly confined optical field in the waveguide by increasing the difference in refractive index. Recently, thin-film lithium niobate (TFLN) prepared on an insulator has attracted intensive attention for use in efficient LN modulators and photonic integrated circuits (PICs). State-of-the-art LN devices that benefit from tight optical-field confinement to realize low-voltage electric modulation and a small footprint include tunable resonators, Mach–Zehnder interferometers (MZIs), and stand-alone phase modulators.^{1,3,6} Extensive research effort has been devoted to developing integrated TFLN devices in PICs.^{8–10} In particular, the heterogeneous integration of TFLN with silicon nitride (SiN)¹¹ or Si^{12,13} rib-loading waveguides will take advantage of mature photonic platforms.¹¹

In early developed TFLN devices, the ridge-shaped waveguide took the upper hand in significant index contrast and high EO confinement. Nevertheless, due to the enormous difference in

permittivity between LN ($\epsilon_{\text{LN}} = 28$ in the Z-axis) and the top cladding ($\epsilon_{\text{air}} = 1.0$, $\epsilon_{\text{SiO}_2} = 3.9$), the electric field is partly dropped on the cladding rather than on the LN ridge, thereby diminishing the divided electric field across the optical mode.¹⁴ In general, a thicker slab and wider ridge are favorable for attaining a lower half-wave voltage (V_π), whereas a looser optical mode will result in propagation loss in the narrow electrode gap. Though TFLN modulators provide more efficient modulation than modulators using bulk LN crystals, some difficulties might exist in the fabrication process. The patterned TFLN waveguide can be prepared using ion-beam etching; however, the low etching selectivity of the TFLN with respect to the mask resist (typically 1:1)¹⁰ is problematic for attaining the precise designed structure. Another problem arises from undesired chemical contamination by nonvolatile by-products in the etching reactor chamber, which is unacceptable for CMOS-compatible processes for constructing electronic integrated circuits. As an approach to resolving this dilemma, a rib-loading waveguide on LN can not only avoid the challenges of the direct etching process but can also retain a strong overlap of the modulated electrical field and optical field. On the basis of this concept, numerous index-matched materials, including tantalum oxide,¹⁵ chalcogenide glass,¹⁶ SiN,¹⁷ and Si,^{12,13} have been used for rib-loading devices. Among them, heterogeneous integration of Si or SiN on modulator devices has attracted intensive interest for use in a CMOS-compatible PIC technology. Some studies have preliminarily demonstrated Si-loaded LN waveguides; however, they show a relatively larger propagation loss (20 dB/cm),¹² low EO modulation efficiency (8.8 V cm), and limited modulation bandwidth (2.5 GHz).¹³ Thus, realizing Si-loaded LN waveguides with an acceptable optical loss, low driving voltage, and wide modulation bandwidth is important and will pave the way for large-scale LN-based PICs in the future.

To fulfill the stringent requirements of on-chip optical integration, a TFLN-based modulator with a low driving voltage and high-speed modulation is regarded as a principal challenge. An efficient TFLN-based modulator with a wide bandwidth can reduce the driving voltage and power consumption, further realizing an on-chip transmitter without a bulky low-noise amplifier. In an attempt to meet this challenge, here we demonstrate a 6.0 mm-long phase-shifter-length heterogeneous Si-on-TFLN (Si/TFLN) EO modulator. A propagation loss of 0.15 ± 0.02 dB/mm, a half-wave voltage length ($V_\pi \cdot L$) product of 1.9 V cm, and an optical signal transmission up to 200 Gbit/s are obtained. To the best of our knowledge, these values are better than those for previously reported Si-loaded LN modulators and are on par with those for state-of-the-art monolithic ridge-shaped waveguide EO modulators.

II. EXPERIMENT AND RESULTS

A. Design and fabrication of the Si/TFLN EO modulator

Schematics showing an overview and cross-section of the Si/TFLN modulator are shown in Figs. 1(a) and 1(b), respectively. A Si core on TFLN with a high refractive index provides lateral optical field confinement along the Y-axis. The width (W_{Si}) and thickness of the Si are 0.5–2.5 μm and 70 nm, respectively. The thicknesses of the TFLN and SiO₂ on the substrate are 700 nm and 2.0 μm , respectively. The input laser is split into two arms by a Y-branch splitter,

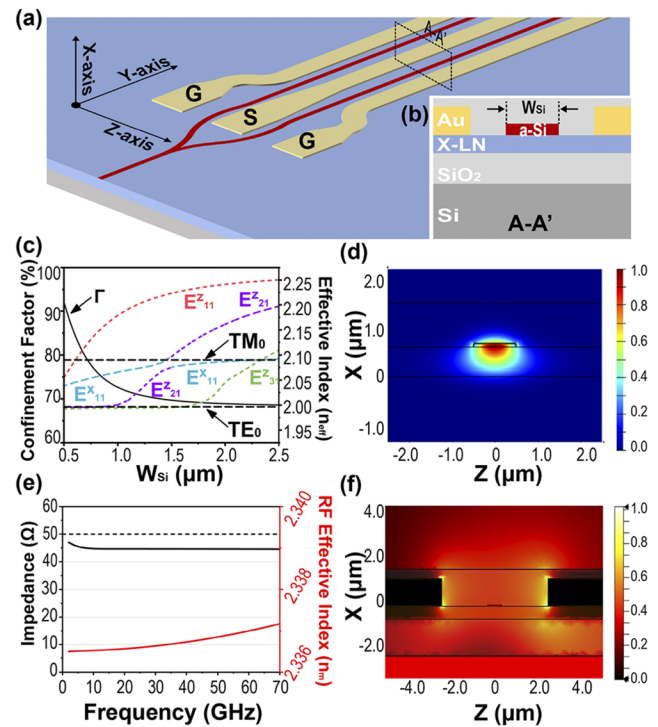


FIG. 1. (a) Schematic of Si/TFLN modulator (Si substrate and upper-cladding SiO₂ not shown). (b) Sectional view of a phase shifter. (c) Calculated confinement factor corresponding to the fundamental TE mode and effective indices for the first four eigenmodes. (d) Simulated optical modal distribution (E_{11}^Z) with 1.0 μm -wide Si. (e) Simulated frequency dependence of impedance and RF effective index. (f) Simulated RF-modulated electric field distribution (E_Z).

and the optical mode size is then compressed using a narrower Si core after 200 μm -long adiabatic tapers. The electrodes have a coplanar ground–signal–ground (GSG) configuration that can apply the opposite electric field in each phase shifter to realize push–pull MZI modulation. A 1.6 μm -thick SiO₂ layer is deposited as the upper cladding. An X-cut TFLN substrate was used in the present study, and the waveguide was designed for the transverse electromagnetic (TE) mode.

In the designed Si/TFLN waveguide, two propagation modes are supported: (1) a mode confined by the Si strip (strip mode), labeled E_{ij}^X and E_{ij}^Z ($i, j = 0, 1, 2, 3, \dots$), where the superscript denotes the direction of the electric vector (X for TM mode, Z for TE mode), and the subscript denotes the mode order, and (2) a mode confined in the TFLN (slab mode), labeled TE_m slab mode and TM_m slab mode ($m = 0, 1, 2, \dots$), where the subscript m denotes the mode order. To avoid leakage loss, the effective refractive index (n_{eff}) for the strip mode should be higher than that for the same-polarization slab mode. Here, we used a mode solver (Rsoft, Synopsys) to calculate the n_{eff} and confinement factor for waveguides with various W_{Si} . The refractive indices for the materials were as follows: $n_{\text{Si}} = 3.475$, $n_{\text{SiO}_2} = 1.444$, $n_{\text{ord,LN}} = 2.211$, and $n_{\text{ext,LN}} = 2.138$, where $n_{\text{ord,LN}}$ and $n_{\text{ext,LN}}$ are the ordinary and extraordinary refractive indices for LN, along the X-axis and Z-axis in Fig. 1(a), respectively.

The effective refractive indices for various Si widths were calculated to dissect the mode evolution of the hybrid waveguide. Figure 1(c) indicates that conversion from E_{11}^X to E_{21}^Z is possible when W_{Si} is increased to $1.5 \mu\text{m}$. To ensure single-mode operation and avoid lateral leakage, the effective index of E_{11}^Z should be higher than that of the TM_0 slab mode, whereas the effective index of E_{21}^Z should be lower than that of the TM_0 slab mode. Thus, W_{Si} should be in the range $0.6\text{--}1.4 \mu\text{m}$. The optical confinement factor (Γ) in this work was defined as the percentage of electromagnetic energy confined within the LN region, as quantified by the following equation:^{2,6}

$$\Gamma = \frac{\iint_{\text{LN}} \text{Re}\{E \times H^*\} \cdot e_z dx dy}{\iint_{\text{total}} \text{Re}\{E \times H^*\} \cdot e_z dx dy}, \quad (1)$$

where E and H are the electric and magnetic field vectors, respectively, and e_z is the unit vector in the Z direction. In Fig. 1(c), the confinement factor corresponding to the E_{11}^Z mode is shown for various W_{Si} . As the width increases, the confinement factor decreases rapidly, which means that the narrower Si is beneficial for obtaining a greater overlap integral between the electric field and the optical mode. Given these facts, we chose $1.0 \mu\text{m}$ -wide Si to ensure low-voltage driving of EO modulation. As depicted in Fig. 1(d), the E_{11}^Z modal profile shows tight optical modal confinement, with an effective mode area of $0.49 \mu\text{m}^2$. Such a relatively small mode area enables the gap between the electrodes to be narrowed without incurring additional electrode absorption loss. Meanwhile, the optical mode is intensely concentrated around the Si/TFLN waveguide. The confinement of the optical mode in the TFLN layer relative to the total modal distribution can be defined by the value of Γ , which is 73%. The simulated optical group index (n_o) for the TE fundamental mode is 2.62, which is similar to the group index for microwave frequency in a dispersive waveguide.¹⁸

The operating bandwidth of the modulator is limited by three factors: impedance mismatch with the external source and load, velocity mismatch, and microwave propagation loss. In the present work, the traveling-wave electrode (TWE) has a thickness and width of 1.5 and $16 \mu\text{m}$, respectively, and a spacing of $6.0 \mu\text{m}$. The signal transmission of the TWE is modeled and numerically calculated using a finite-difference time-domain solver (CST Studio Suite). As shown in Fig. 1(e), the simulated impedance is $\sim 44 \Omega$ at frequencies as high as 70 GHz . In addition, the radio-frequency (RF) effective index (n_m) is evaluated at around 2.34, which is smaller than the value of 6.0 for a bulk LN crystal and can ease the restriction of the bandwidth caused by the velocity mismatch between light and RF waves.¹⁹ Assuming negligible microwave propagation loss, the 3 dB bandwidth-length product ($f_{3\text{dB}}L$) can be estimated as $1.4c/(\pi|n_o - n_m|)$ where c is the speed of light in vacuum.²⁰ In this design, the theoretical $f_{3\text{dB}}L$ product can be estimated to be 47.8 GHz cm , corresponding to a modulation frequency limit of 79.7 GHz for a 6.0-mm -long transmission line. Consequently, the RF propagation loss, which is determined by the conduction loss and dielectric loss, in the TWE is expected to be the limiting factor for the bandwidth. The simulated RF electric field between electrodes is presented in Fig. 1(f), which shows a uniform electric-field distribution in the TFLN between electrodes. There exists little

electric field discontinuity at the boundaries between the Si, TFLN, and SiO_2 cladding. Therefore, the maximum modulation efficiency can be expected from the given electric field in the Si/TFLN modulator.

Device fabrication started with a TFLN wafer consisting of a 700 nm -thick LN layer on a $2.0 \mu\text{m}$ -thick SiO_2/Si substrate (NANOLN). A 70 nm -thick layer of amorphous Si (a-Si) was deposited onto the TFLN by plasma-enhanced chemical vapor deposition (PECVD) with SiH_4 . A low-temperature PECVD process at 150°C was used to deposit an optically transmissive a-Si layer with a precise thickness and a low surface roughness while avoiding cracking and peeling caused by the different thermal expansion coefficients of the materials. The waveguide was patterned on Si using a standard electron-beam lithography technique and reactive-ion etching with SF_6 . The coplanar electrodes, which were made of 10 nm -thick Cr and $1.5 \mu\text{m}$ -thick Au, were fabricated via a lift-off and metal vacuum deposition technique. Subsequently, a $1.6 \mu\text{m}$ -thick SiO_2 cladding layer was deposited on top by CVD. The electrode contact pads were opened using a buffered oxide etching solution. The devices were finally diced and end-face-coupled to high-NA single-mode fibers.

B. Device characterization

Figure 2 shows optical microscopy images of the fabricated Si/TFLN modulator. The Si core was $1.0 \mu\text{m}$ wide. The length of the phase shifter necessary to achieve an acceptable driving voltage was 6.0 mm , which is substantially shorter than that for previous LN modulators. The 3 dB power splitter and coupler consisting of a symmetric Y-branch were designed to obtain a balanced MZI configuration and have a total length of 0.8 mm to avoid bending loss. A $200\text{-}\mu\text{m}$ -long adiabatic taper was used to connect the waveguide to the phase shifter. To evaluate the propagation loss for the Si/TFLN structure, waveguides having different lengths (cut-back) were fabricated, and their optical transmission was measured at a wavelength of 1550 nm . The results indicated a propagation loss of $0.15 \pm 0.02 \text{ dB/mm}$. Optical transmission measurements of the Si/TFLN modulator were performed to characterize the EO modulator. Light from a tunable laser was conducted to the end face of the fabricated waveguide via a polarization-maintaining fiber. The wavelength was set at 1550 nm . The output light was conducted to a photodetector. The on-chip loss for the fabricated device was estimated to be 2.7 dB , comprising a phase-shifter loss of 0.9 dB and an other passive component loss of 1.8 dB . The additional fiber-to-chip coupling loss was estimated to be $\sim 4 \text{ dB/facet}$, which can be reduced by using, for example, a standard fiber-to-PIC coupling strategy.²¹

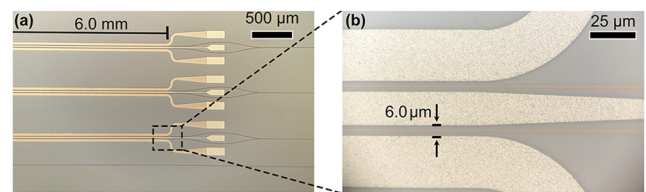


FIG. 2. (a) Microscopy image of the MZI structure of the Si/TFLN modulator. (b) Magnified microscopy image of area between electrodes in (a).

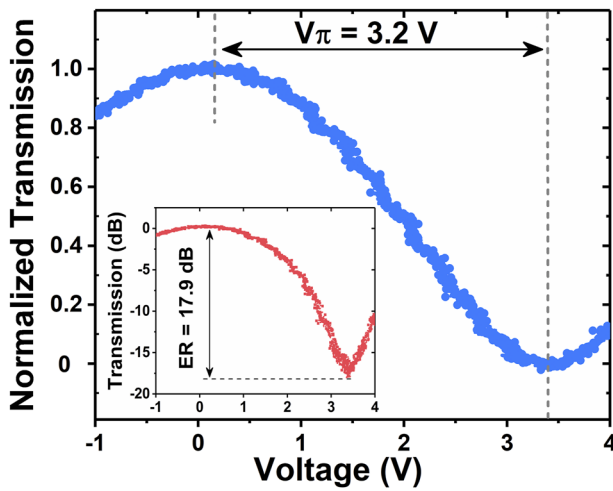


FIG. 3. Normalized transmission for the Si/TFLN modulator with the 6.0 mm-long phase-shifter MZI under various drive voltages. Inset: measured transmission (dB) normalized to maximum received power.

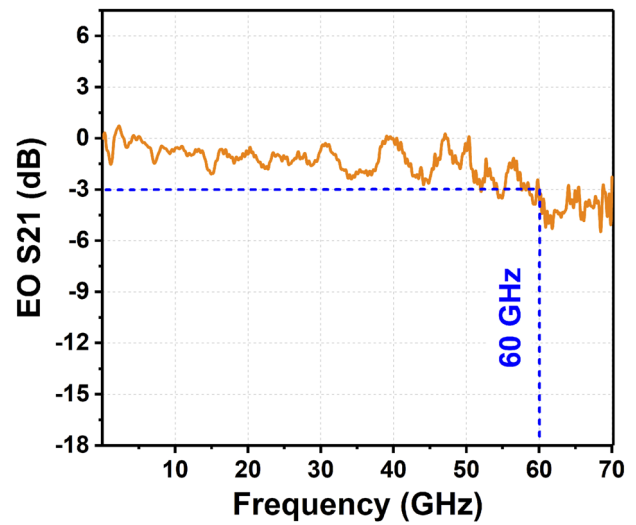


FIG. 4. EO frequency response (EO S21) of the Si/TFLN modulator.

The fabricated MZI modulator consisted of a balanced interferometer with same-length phase shifters and coplanar GSG electrodes. Thus, conventional electrical push-pull driving resulted in efficient EO modulation. A triangle-wave electric signal with a frequency of 10 kHz and a voltage of 5.0 V_{pp} was applied to the modulator. To measure the V_π , the output optical signal was detected using an amplified photodetector and recorded along with the input electrical voltage using an oscilloscope. As shown in Fig. 3, the measured V_π was 3.2 V, corresponding to a $V_\pi \cdot L$ of 1.9 V cm when a phase-shifter length of 6.0 mm is considered. This product value is nearly one order of magnitude lower than the values reported for previous bulk LN modulators.²² Even compared with the state-of-the-art TFLN modulator,⁸ a further 18% reduction of the $V_\pi \cdot L$ can be realized in this modulator. The in-device EO coefficient (r_{eff}) can be calculated using the following equation under the assumption of an ideal uniform electric-field distribution:

$$r_{\text{eff}} = \frac{\lambda g}{2n^3 \Gamma V_\pi L}, \quad (2)$$

where λ is the wavelength, g is the electrode gap, and n is the effective refractive index of the waveguide. The calculation result agrees with our electric field simulation result and provides an in-device EO coefficient of around 31 pm/V, which is consistent with the attainable EO coefficient for LN. The measured static extinction ratio (ER) was 17.9 dB, which is comparable to the value reported for other heterogeneous TFLN modulators.^{16,23} Notably, a much higher ER can be obtained by improving the balanced splitters after improvement of the splitter loss or by using a multimode interferometer splitter.²⁴

C. Data transmission performance

The EO modulation bandwidth was characterized using a vector network analyzer (MS4647B, Anritsu) in the frequency range from 0.01 to 70 GHz, and the results are shown in Fig. 4. Small

RF signals were supplied to the electrode of the modulator using an RF probe (GSG AS65, Form Factor). Another RF probe with a 50 Ω termination was attached to the end of the electrode to reduce RF reflection. The modulated signal was received by a calibrated photodetector (MN4765B O/E calibration module, Anritsu). The measurement was calibrated using an automatic calibrator prior to use (36585 Auto Cal, Anritsu). The EO response (S21) exhibits a 3 dB bandwidth of 60 GHz. According to the theoretic

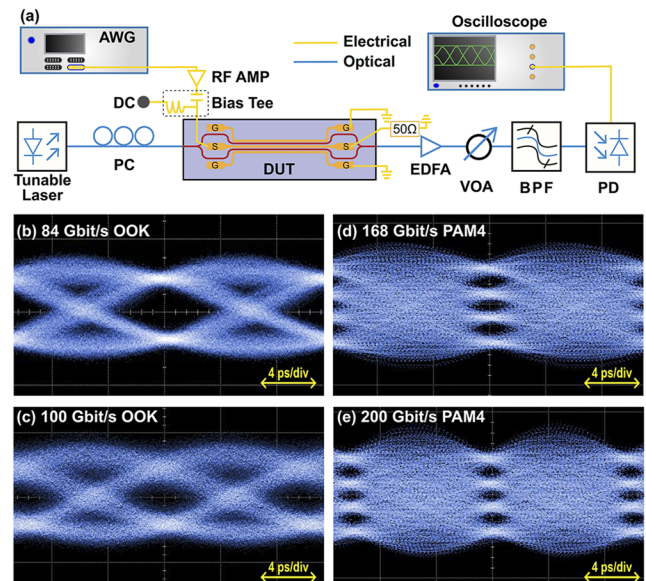


FIG. 5. High-speed signal generation by Si/TFLN modulator. (a) Experimental setup for signal generation and detection. Detected eye patterns for (b) 84 Gbit/s and (c) 100 Gbit/s OOK signals and (d) 168 Gbit/s and (e) 200 Gbit/s PAM4 signals.

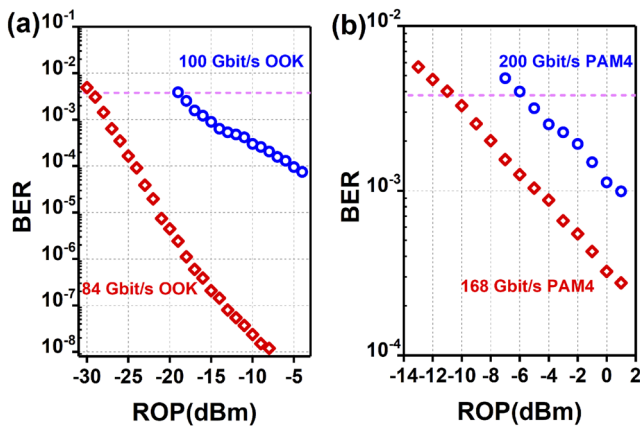


FIG. 6. BER vs received optical power (ROP). (a) BER for 84 and 100 Gbit/s OOK. (b) BER for 168 and 200 Gbit/s PAM4. Pink dashed lines: 7% HD-FEC threshold of 3.8×10^{-3} .

cal model discussed in Sec. II A, a higher frequency response can be expected for the designed Si/TFLN device. If the phase shifter can be experimentally improved and fabrication-related imperfections can be reduced, increasing the 3 dB bandwidth to close to the velocity mismatch bandwidth limit of 79.7 GHz might be possible.

To evaluate the feasibility of the Si/TFLN modulator for high-data-rate operation, we applied an electrical signal and verified the modulation rate at data transfer speeds as high as 200 Gbit/s. In the experimental setup shown in Fig. 5(a), a $2^{11}-1$ pseudo-random binary sequence data stream was output by an arbitrary waveform generator (AWG M8194A, Keysight) and then amplified by a linear broadband amplifier (S804B, SHF) to $1.8 V_{pp}$. The optical signal was received by a 70 GHz signal photodetector (PD XPDV3120R, Finisar) after passing through an Er-doped fiber amplifier (EDFA) and a band-pass filter (BPF) and then fed to a digital communication analyzer (DCA86116C, Keysight) for eye-pattern generation. Figures 5(b) and 5(c) show the eye patterns measured for 84 and

100 Gbit/s on-off keying (OOK), respectively. A linear feed-forward filter was applied for equalization (tap = 4) over the digitized data in the analysis. The measured eye patterns are clear in the present work, with a Q-factor of 7.2 for 84 Gbit/s and 5.3 for 100 Gbit/s, for which successful transmission is expected. Extinction ratios are extracted from measured eye patterns as 3.8 and 3.0 dB for 84 and 100 Gbit/s, respectively. To further assess the modulation performance, the bit error rate (BER) was analyzed using a real-time oscilloscope (UXR1102A, Keysight) for OOK signals with offline post-processing that included timing recovery, feed-forward equalization, and error counting. The measured BERs at signaling rates of 84 and 100 Gbit/s are presented in Fig. 6(a), where the power of the light before the PD was adjusted. To obtain receiver sensitivity curves, a variable optical attenuator (VOA) was placed in front of the PD, and the BER change vs the received optical power (ROP) was analyzed. For analysis purposes, the 7% hard-decision forward error correction (HD-FEC) threshold (3.8×10^{-3}) was used as a reference. The Si/TFLN modulator achieves a BER level of 1.2×10^{-8} at 84 Gbit/s with an ROP of -8 dBm. The sensitivity curve increased by several orders of magnitude at 100 Gbit/s. However, the BER was recorded as 7.5×10^{-5} at an ROP of -4 dBm and increased to the level of the 7% HD-FEC limit at -19 dBm.

The EO linear response based on the LN crystal originates from the Pockels effect, which makes the modulators suitable for multilevel modulation such as four-level pulse-amplitude modulation (PAM4). In PAM4, the modulator maintains the highest baud rate while doubling the signal transmission rate. Thus, we applied PAM4 signals at 168 and 200 Gbit/s to the Si/TFLN modulator. The measured PAM4 eye patterns are shown in Figs. 5(d) and 5(e). A clear eye opening was obtained after the feed-forward equalizer. The Nyquist shaping of the signal was tuned by applying a small roll-off factor of 0.4 in the raised cosine filter, which could further boost the transmission of the modulator with enhanced signaling efficiency. The BER analysis revealed low BERs of 2.6×10^{-4} at 168 Gbit/s and 1.0×10^{-3} at 200 Gbit/s.

To assess the receiver sensitivity, the measured BER curves for 168 and 200 Gbit/s at various ROPs are shown in Fig. 6(b). For the 200 Gbit/s PAM4, we measured error-free signals when the ROP was above -6 dBm. An approximately two-thirds decrease in the

TABLE I. Comparison of performance metrics for LN-based EO modulators.

Devices	$V_{\pi} \cdot L$ (V·cm)	Length of active section (mm)	EO S_{21} roll-off	Line rate (Gbit/s)	LN etching free	References
Si-loading	1.90	6.0	-3 dB (at 60 GHz)	200 (PAM4)	Yes	This work
Si-loading	8.8	0.8	-3 dB (at 2.5 GHz)	4.5 (OOK)	Yes	13
SiN-loading	2.24	7.8	-3 dB (at 30 GHz)	80 (OOK)	Yes	23
SiN-loading	2.11	24	-3 dB (at 29 GHz) ^a	N/A	Yes	17
TFLN on SOI	6.70	5.0	-3 dB (at 106 GHz)	20 (OOK)	Yes	26
Etched LN on SOI	2.20	5.0	-3 dB (at 70 GHz)	112 (PAM4)	No	27
Monolithic	2.20	10.0	-1.4 dB (at 67 GHz)	112 (PAM4)	No	28
Monolithic	2.80	20.0	-3 dB (at 45 GHz)	210 (8-ASK)	No	3
Legacy Ti: LN	>14	>40 ^a	-3 dB (at 25 GHz)	40 (OOK)	Yes	29
Legacy Ti: LN	>20	>40 ^a	-3 dB (at 30 GHz)	40 (OOK)	Yes	30

^aEstimated from the reference.

BER was measured for 168 Gbit/s PAM4. In our PAM4 experiments at such high baud rates, electric driving of the modulator was conducted with priority given to the lower BER signals; thus, a Nyquist filter was applied. At a given symbol rate, the power penalty between the OOK and PAM4 formats can be expected to be 4.8 dB ($=10 \log 3$).²⁵ However, an excess penalty in the measured result is observed, which is partly attributed to the digital signaling processes with different Nyquist roll-off factors and equalizations. The amplitude control of the driver and bias voltages can be used to more precisely tune the PAM4 signals. The adjustments are currently done manually; both inaccuracy and drift improvements would increase the signal accuracy in PAM4 transmission.

Finally, Table I shows a comparison between the values of several performance metrics obtained in the present work and those reported for state-of-the-art results reported in recent years using different waveguide types of LN-based EO modulators. We highlight the fact that the present work is the first demonstration of 200 Gbit/s PAM4 transmission with a BER level lower than the HD-FEC threshold (7%). Based on the expected EO coefficient for LN, the minimum $V_{\pi} \cdot L$ can be estimated to be 1.9 V cm, as demonstrated in the present work. The measured $V_{\pi} \cdot L$ agrees with our design principle, which focuses on efficient modulation.

III. CONCLUSION

We reported the fabrication of an LN-etching-free Si/TFLN waveguide and its application in an MZI modulator. The proposed EO modulator exhibited a $V_{\pi} \cdot L$ product of 1.9 V cm and a large extinction ratio of 17.9 dB. Clear eye patterns were experimentally demonstrated at data rates as high as 100 Gbit/s in OOK and 200 Gbit/s in PAM4 with BER values below the HD-FEC threshold (7%). Compared with previous state-of-the-art LN devices, the proposed Si/TFLN modulator has a straightforward fabrication process and demonstrates higher bit-rate transmission performance, providing a potentially lower-cost and CMOS-compatible solution for large-scale integrated photonic circuits.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jiawei Mao: Conceptualization (equal); Investigation (equal); Writing – original draft (equal). **Hiromu Sato:** Conceptualization

(equal); Investigation (equal); Writing – original draft (equal). **Guo-Wei Lu:** Conceptualization (equal); Investigation (equal); Writing – original draft (equal). **Shiyoshi Yokoyama:** Conceptualization (equal); Funding acquisition (lead); Investigation (equal); Project administration (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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